Intro to CMOS Processing

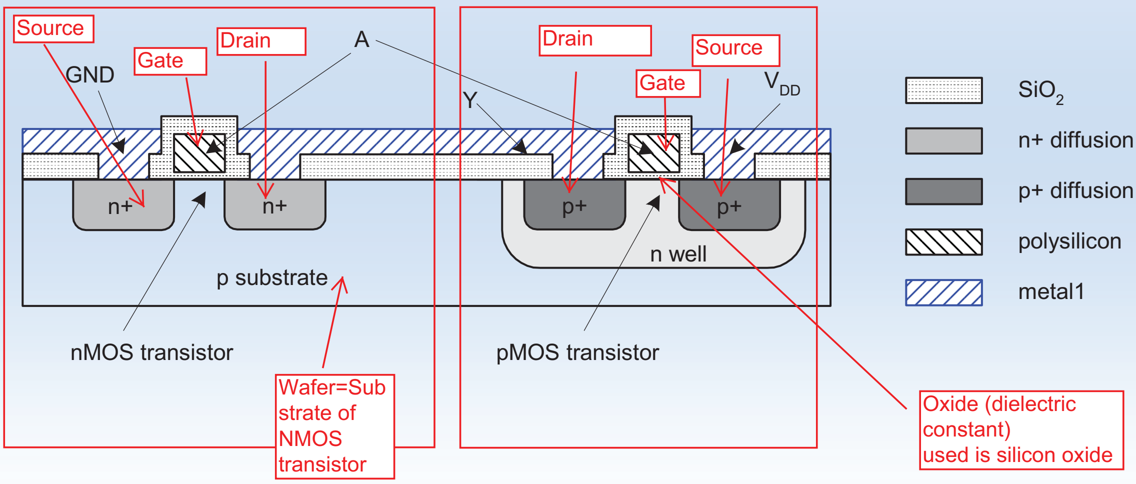
- CMOS transistors are fabricated on silicon wafer
- Lithography process similar to printing press
- On each step, different materials are deposited or etched
- Easiest to understand by viewing both top and cross-section of wafer in a simplified manufacturing process

Note: **Slides are taken from**

Weste: CMOS VLSI Design, Addison Wesley, 2004

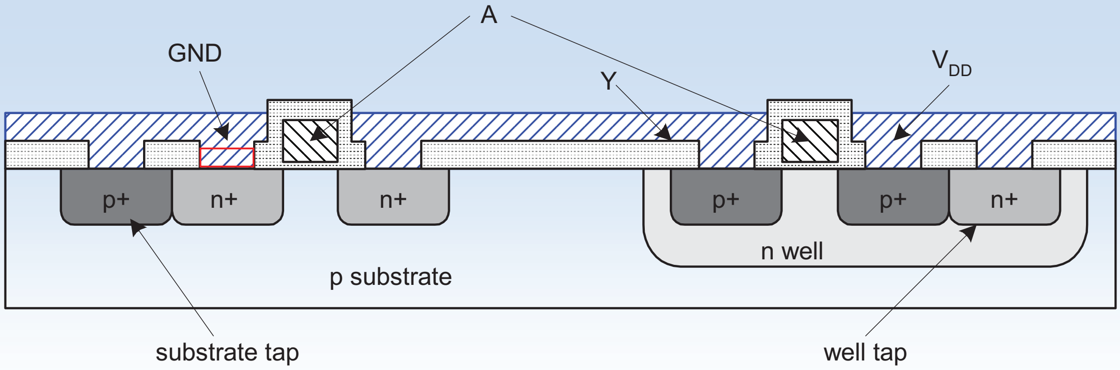
Inverter Cross-section

- Typically use p-type substrate for nMOS transistors
- Requires n-well for body of pMOS transistors



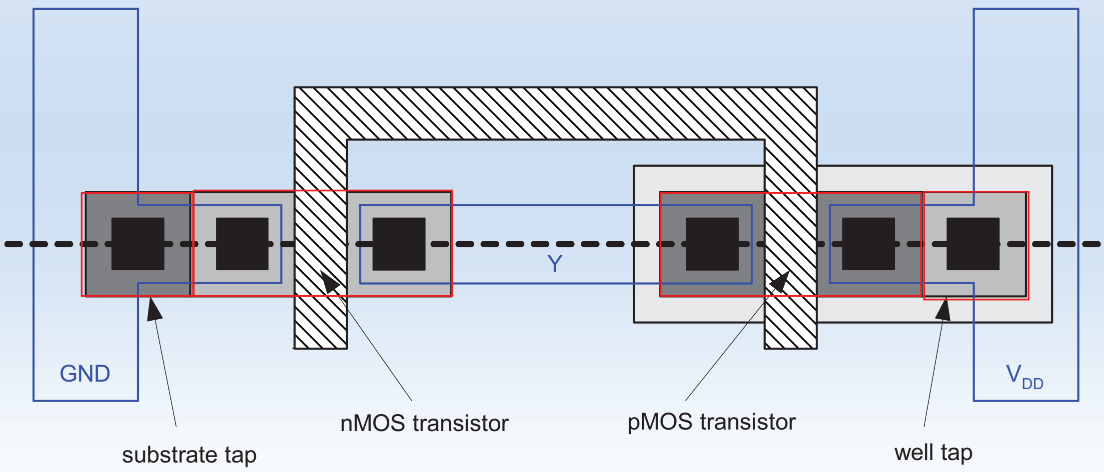
Well and Substrate Taps

- Substrate must be tied to GND and n-well to V_{DD}
- Use heavily doped well and substrate contacts / taps



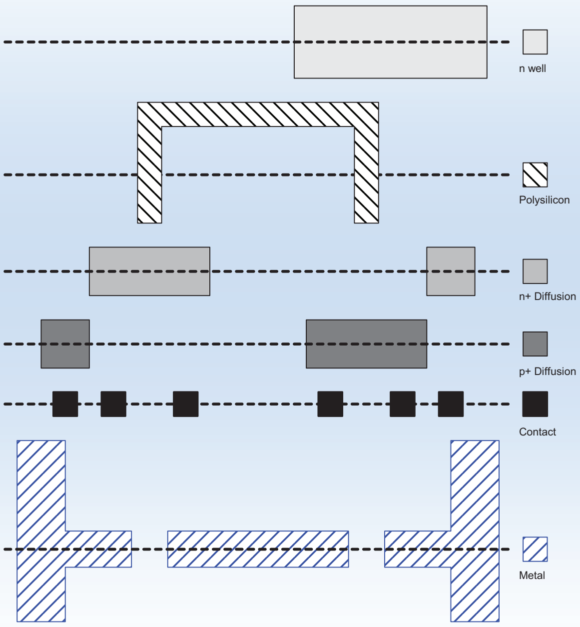
Inverter Mask Set

- Transistors and wires are defined by *masks*
- Cross-section taken along dashed line on previous slide



Detailed Mask Views

- Six masks
  - n-well
  - Polysilicon
  - n+ diffusion
  - p+ diffusion
  - Contact
  - Metal

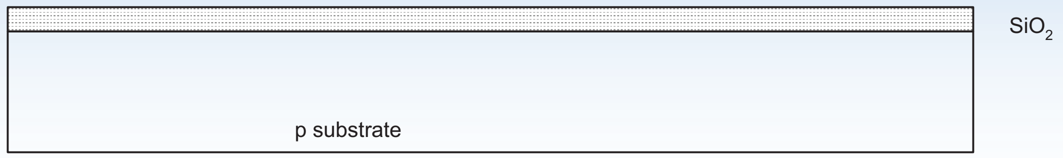


Fabrication Steps

- Start with blank wafer
- Build inverter from the bottom up
- First step will be to form the n-well
  - Cover wafer with protective layer of SiO_2 (oxide)
  - Remove layer where n-well should be built
  - Implant or diffuse n dopants into exposed wafer
  - Strip off SiO_2

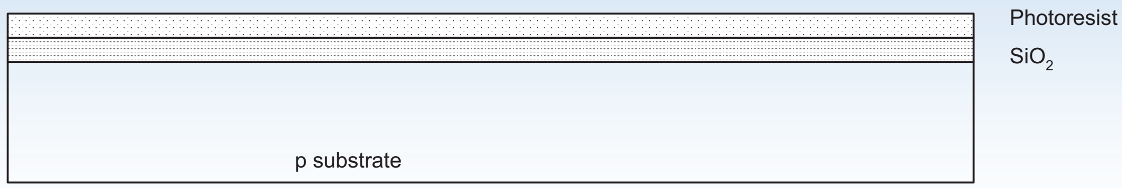
Oxidation

- Grow SiO_2 on top of Si wafer
  - 900 – 1200 C with H_2O or O_2 in oxidation furnace



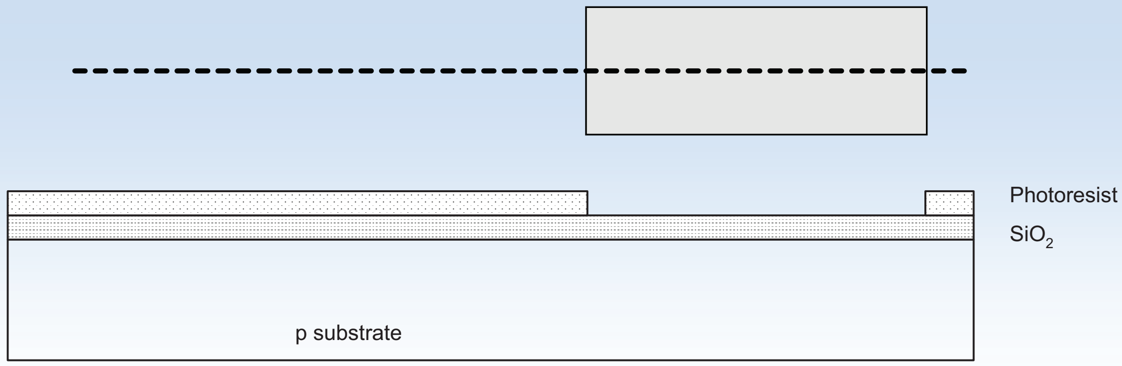
Photoresist

- Spin on photoresist
  - Photoresist is a light-sensitive organic polymer
  - Softens where exposed to light



Lithography

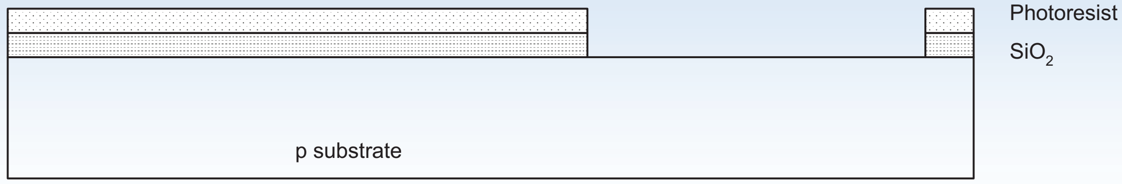
- Expose photoresist through n-well mask
- Strip off exposed photoresist



IC Fabrication

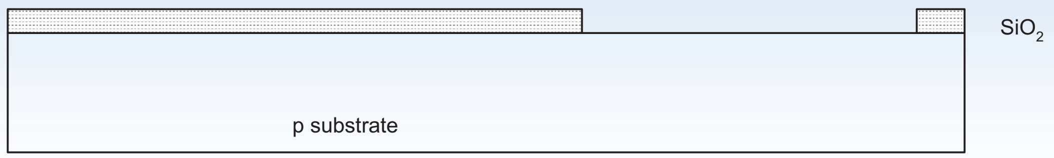
Etch

- Etch oxide with hydrofluoric acid (HF)
  - Seeps through skin and eats bone; nasty stuff!!!
- Only attacks oxide where resist has been exposed



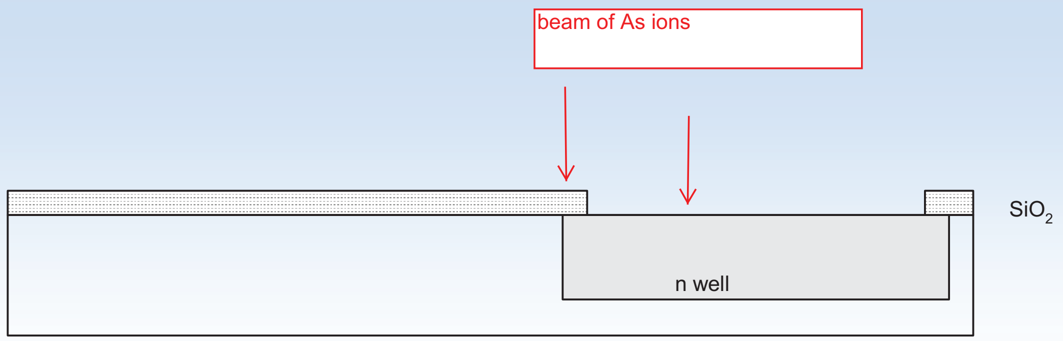
Strip Photoresist

- Strip off remaining photoresist
  - Use mixture of acids called piranah etch
- Necessary so resist doesn't melt in next step



n-well

- n-well is formed with ion implantation
- Ion Implantation
  - Blast wafer with beam of As ions
  - Ions blocked by SiO_2, only enter exposed Si

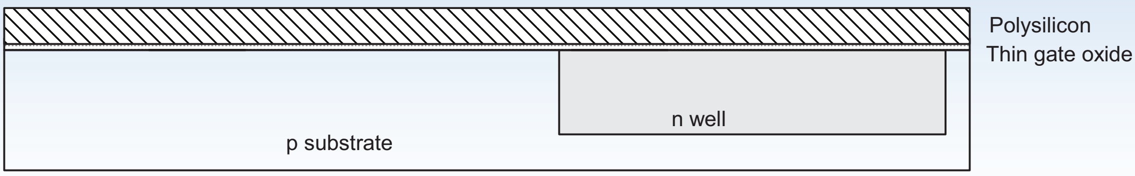


Strip Oxide

- Strip off the remaining oxide using HF
- Back to bare wafer with n-well
- Subsequent steps involve similar series of steps

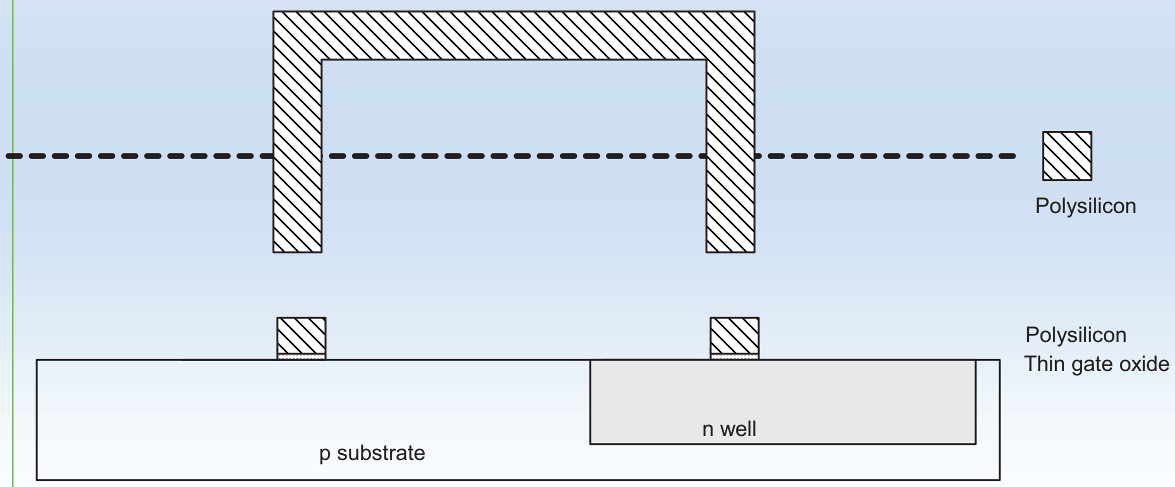
Polysilicon

- Deposit very thin layer of gate oxide
  - < 20 Å (6-7 atomic layers)
- Chemical Vapor Deposition (CVD) of silicon layer
  - Place wafer in furnace with Silane gas (SiH_4)
  - Forms many small crystals called polysilicon
  - Heavily doped to be good conductor



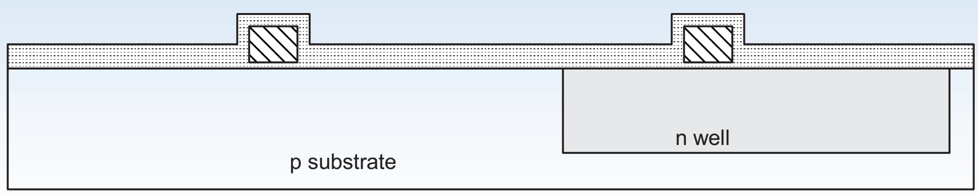
Polysilicon Patterning

- Use same lithography process to pattern polysilicon



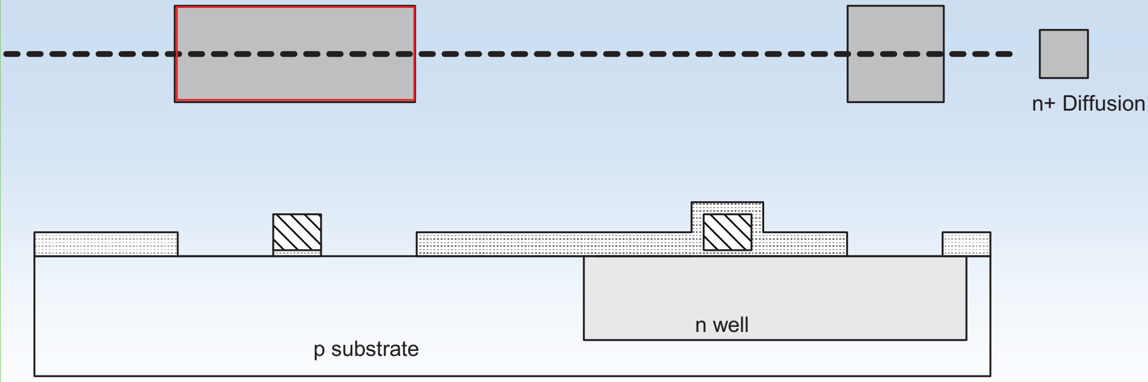
Self-Aligned Process

- Use oxide and masking to expose where n+ dopants should be diffused or implanted
- N-diffusion forms nMOS source, drain, and n-well contact



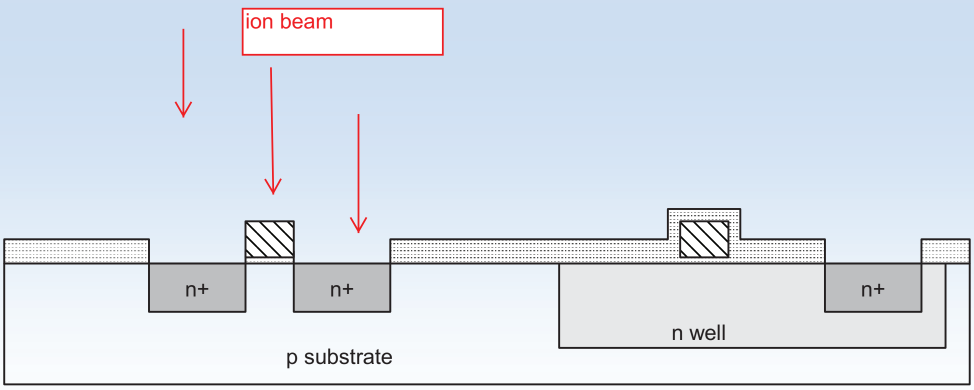
N-diffusion

- Pattern oxide and form n+ regions
- *Self-aligned process* where gate blocks diffusion
- Polysilicon is better than metal for self-aligned gates because it doesn't melt during later processing



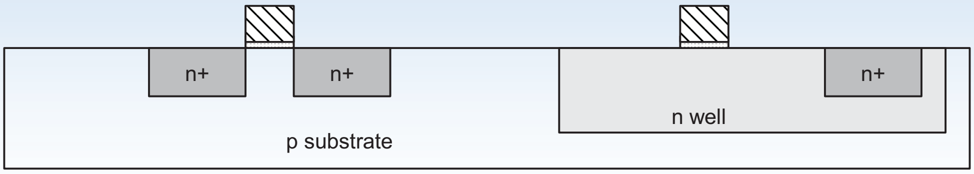
N-diffusion cont.

- Historically dopants were diffused
- Usually ion implantation today
- But regions are still called diffusion



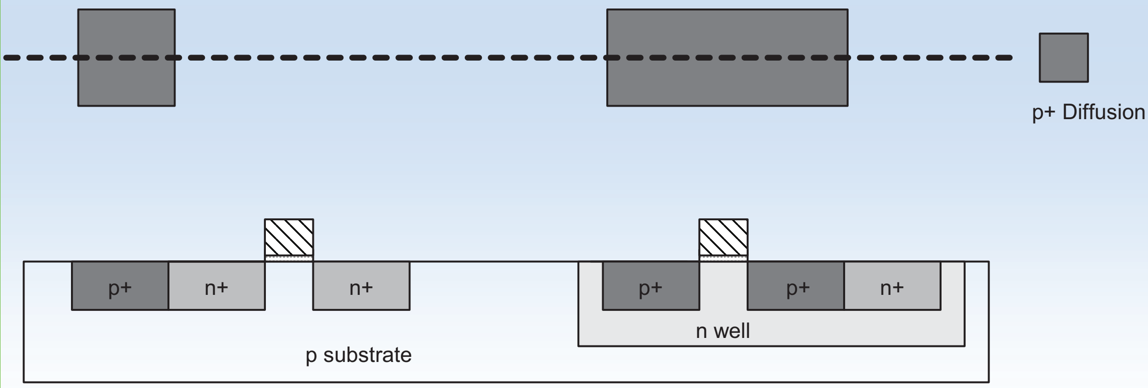
N-diffusion cont.

- Strip off oxide to complete patterning step



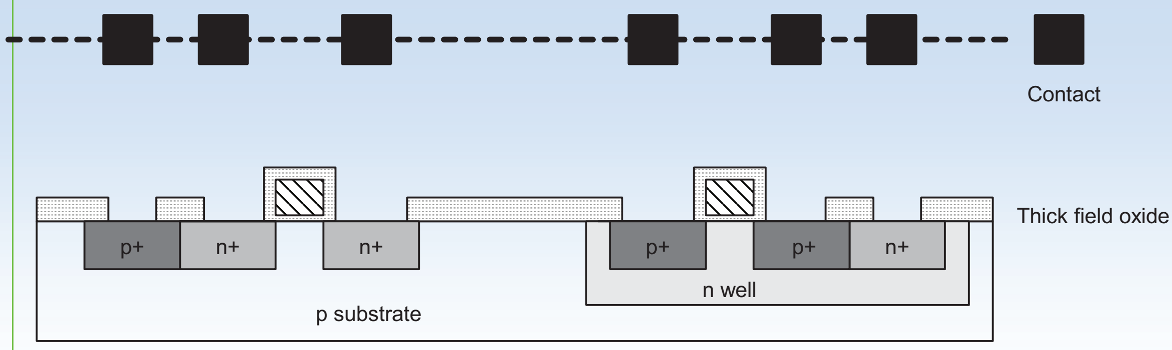
P-Diffusion

- Similar set of steps form p+ diffusion regions for pMOS source and drain and substrate contact



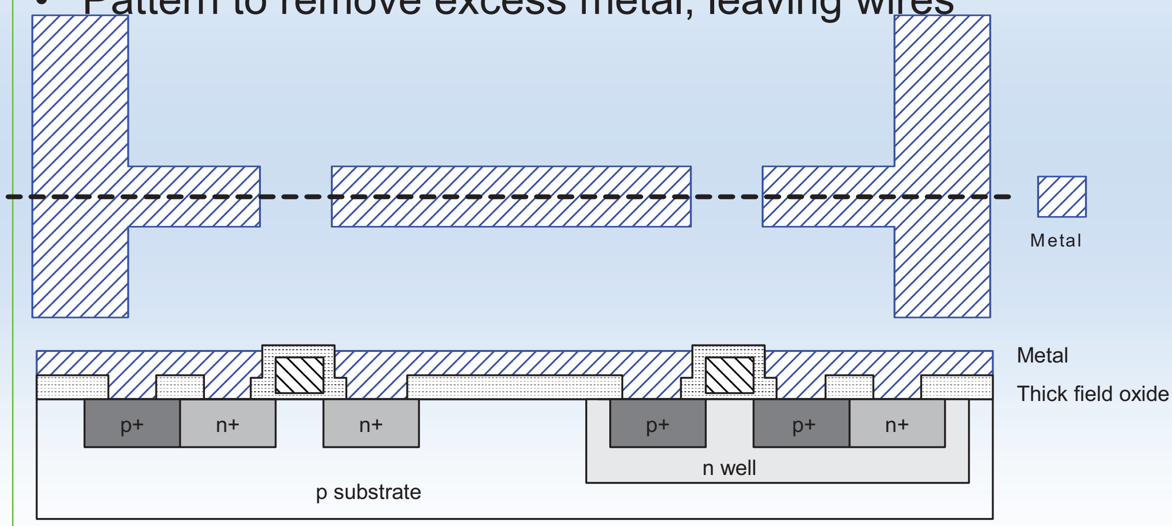
Contacts

- Now we need to wire together the devices
- Cover chip with thick field oxide
- Etch oxide where contact cuts are needed



Metalization

- Sputter on aluminum over whole wafer
- Pattern to remove excess metal, leaving wires

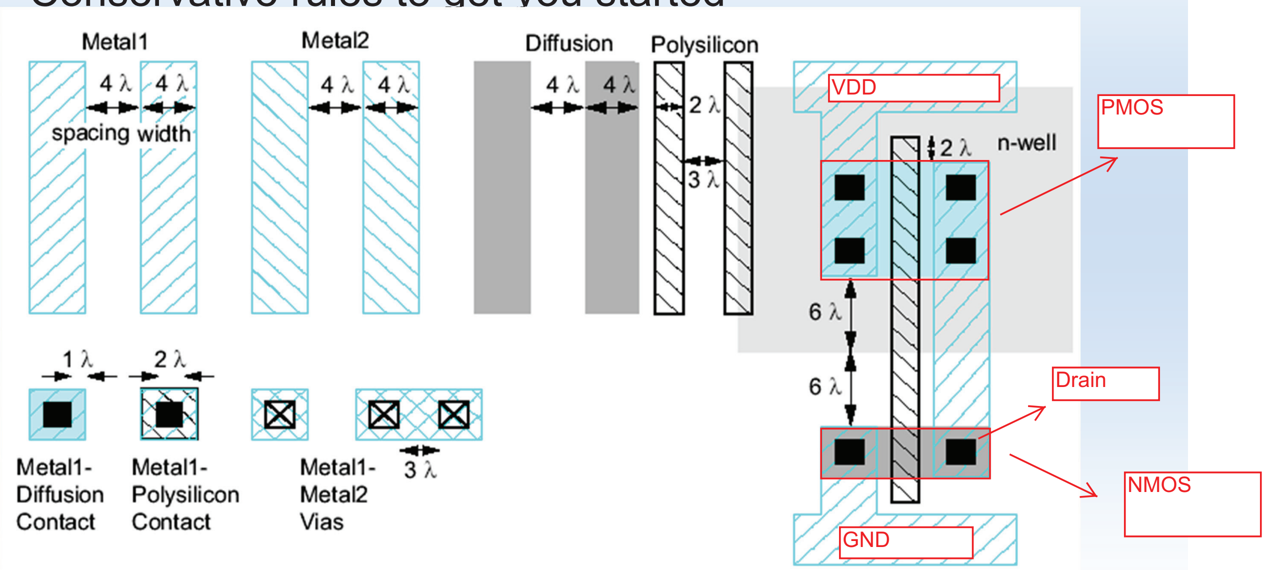


Layout

- Chips are specified with set of masks
- Minimum dimensions of masks determine transistor size (and hence speed, cost, and power)
- Feature size f = distance between source and drain
  - Set by minimum width of polysilicon
- Feature size improves 30% every 2-3 years
- Normalize for feature size when describing design rules
- Express rules in terms of $\lambda = f/2$
  - E.g. λ = 5 nm in 10 nm CMOS process

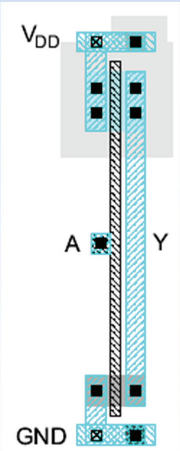
Simplified Design Rules

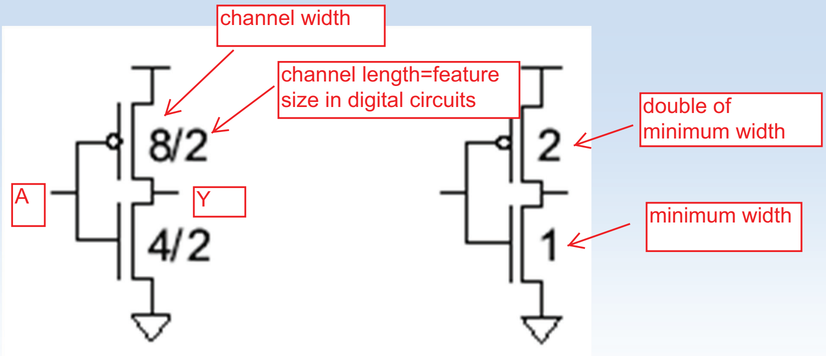
- Conservative rules to get you started



Inverter Layout

- Transistor dimensions specified as Width / Length
  example: minimum size is 4λ / 2λ



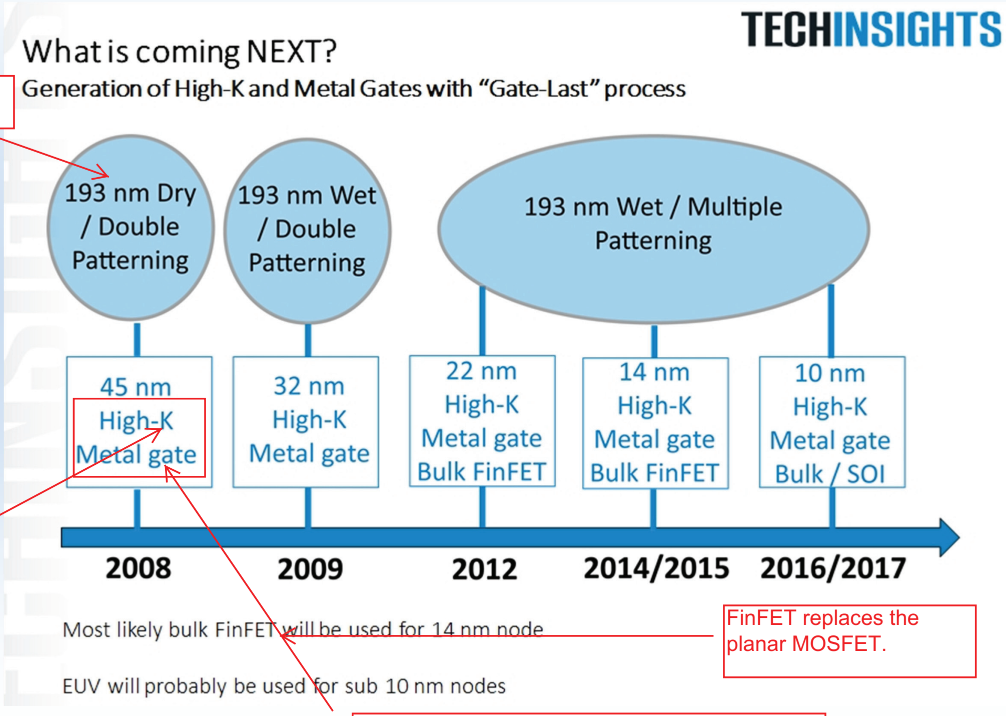


Summary

- Masks specify the dimension and the location of transistor structures and wires
- For patterning the lithography process is used
- On each step of the chip fabrication different materials are deposited or etched away
- To build the source and drain of the MOS-transistors a self-aligned lithography process is used

High-k/metal gate/FinFET

Lithography wave length

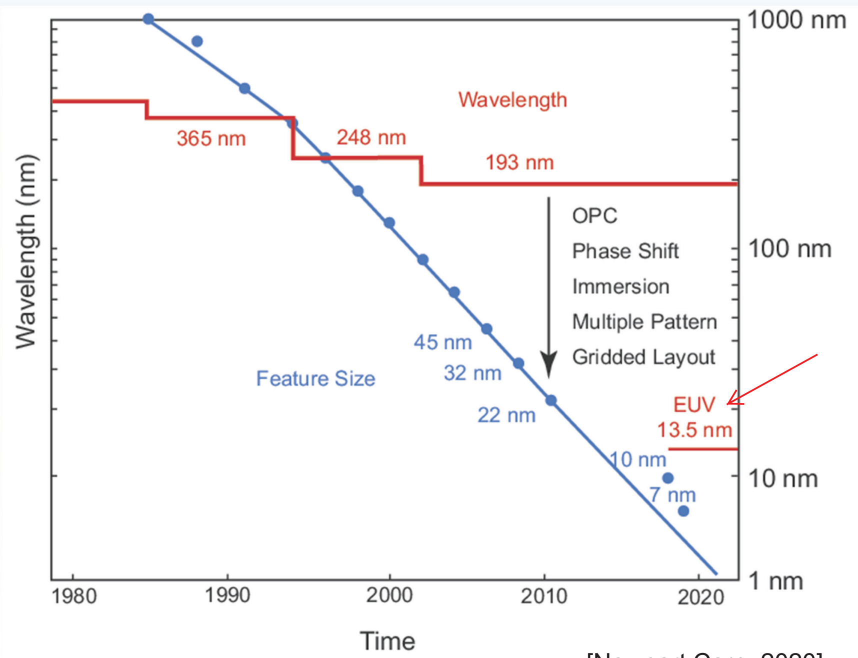


high dielectric constant advantage:
1. higher drive current
2. reduced gate leakage current (reduced power consumption)

FinFET replaces the planar MOSFET.

replacing polysilicon by a metal advantage:
improved conductivity results in higher signal speed

Photolithography Technology



[Newport Corp. 2020]

VLSI Economics

- Selling price S_{total}

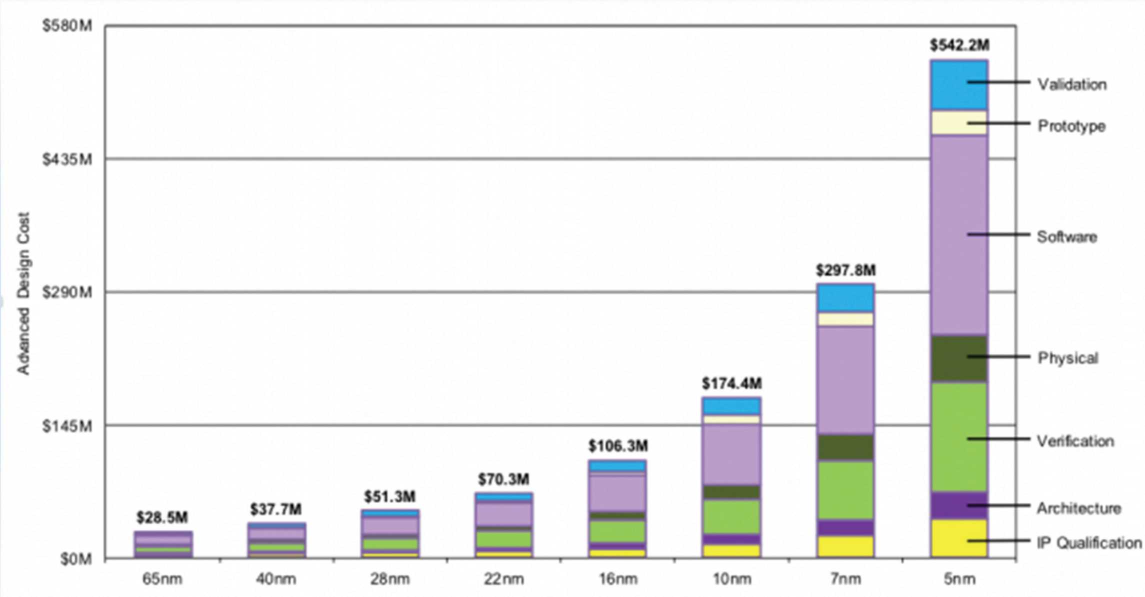
  $S_{total} = C_{total} / (1-m)$

- m = profit margin
- C_{total} = total cost
  - Nonrecurring engineering cost (NRE)
  - Recurring cost
  - Fixed cost

VLSI Economics:NRE

- R&D Engineering cost
  - payroll costs
  - office & workstations
  - EDA tools
- Prototype manufacturing

VLSI Economics:NRE



[IBS 2018]

12/14/16nm chips: cost of roughly $100M to design a new GPU, CPU, or SoC. But with 7nm design cost has tripled.

VLSI Economics:Recurring Cost

- ❑Fabrication

$$\text{cost of one die} = \frac{\text{wafer cost}}{\text{total number of dies per wafer} \cdot \text{yield}}$$

Number of dies per wafer:

A=Area of die (chip)

r=radius of wafer

$$N = \pi \left[\frac{r^2}{A} - \frac{2r}{\sqrt{2A}} \right]$$

Yield: $$Y = \frac{\text{number of functional dies per wafer}}{\text{total number of dies per wafer}}$$

VLSI Economics:Recurring Cost

❑Fabrication

Example (5nm CMOS process, 2020):

wafer cost=17000$

wafer diameter=300mm, chip area=610mm^2

yield=90%

Q: fabrication cost of one die?

number of dies per wafer N=88, cost of one die = $\frac{17000\$}{88 \times 0.9}$=215$

❑Packaging

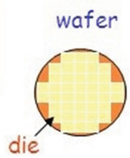
❑Test

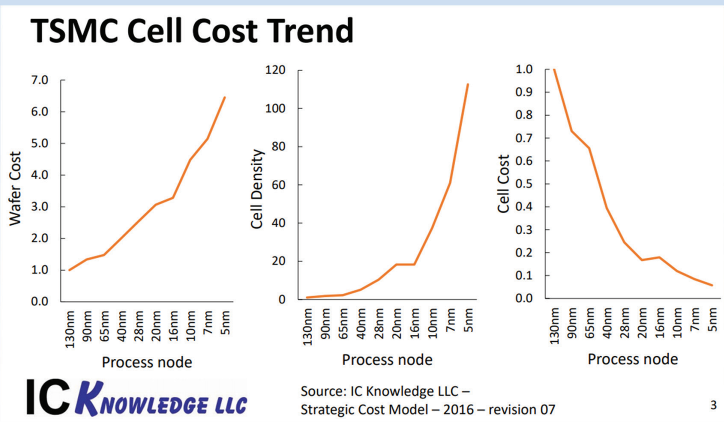
VLSI Economics:Recurring Cost

❑Total IC Cost

$$\text{IC Cost} = \frac{\text{Die Cost} + \text{Testing Cost} + \text{Packaging Cost}}{\text{Final Test Yield}}$$

$$\text{Die Cost} = \frac{\text{Wafer Cost}}{(\text{Dies/Wafer}) \times \text{Die Yield}}$$





VLSI Economics:Fixed Costs

- Data sheets and application notes
- Marketing and advertising
- Yield analysis